

Rates of local control, metastasis, and overall survival in patients with posterior uveal melanomas treated with ruthenium-106 plaques

Markus Kleineidam^a, Rudolf Guthoff^b, Søren M. Bentzen^{*c}

^a*Institute of Biophysics and Radiobiology, ^bDepartment of Ophthalmology, University Clinic Hamburg Eppendorf, University of Hamburg, Hamburg, Germany*

^c*Danish Cancer Society, Department of Experimental Clinical Oncology, Nørrebrogade 44, DK-8000 Aarhus C, Denmark*

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Abstract

The prognostic importance of clinically assessed tumor variables, patient variables, and treatment variables of 184 patients treated with ruthenium-106 eye plaques for uveal melanoma was studied using crude survival, metastatic disease and local tumor control as the endpoints. Using the multivariate Cox proportional hazards model, tumor diameter, age and sex were identified as the strongest variables predicting survival. Risk of metastasis was found to be dependent on tumor size (diameter) only and the metastasis was generally observed within 6 years after the end of treatment (96%). Clinically diagnosed local tumor control was predicted by tumor T-stage, treatment time, and total dose to the sclera. No correlation between metastatic disease and local failure was detected. However, the entanglement of tumor size and treatment characteristics prevents really firm conclusions being made on optimal therapy, but the results suggest that: (i) the rate of survival and the rate of metastasis after some years might be poor endpoints to compare different treatment modalities, since probably most tumors have already metastasized prior to treatment; (ii) patients with re-treated tumors do not have a considerably worse prognosis than those who were treated successfully; (iii) tumors of more than 10 mm in diameter should be considered to be a systemic disease and interest should focus on possible adjuvant systemic therapies; (iv) a prognostic index of tumor variables prior to treatment has to be taken into consideration to compare different treatment modalities; (v) there is no clear therapeutic benefit from using plaques of high dose-rate concerning local control and severe side-effects; and (vi) sex and age have no impact on the disease related death.

Key words: Uveal melanoma; Metastases; Brachytherapy; Ruthenium-106

1. Introduction

While many studies have tried to explore factors that predict whether an uveal melanoma will metastasize or not [2,7,9,12,24,26,27,38,45,48–50], few have considered the influence of radiobiological parameters on the outcome of radiotherapy [14,41]. Some tumor and patient characteristics are considered to have prognostic value such as tumor volume, largest tumor diameter, age, sex, histopathology [25,31], location, invasion, pigmentation and time to regression [1,5,24,31].

Nevertheless, optimal therapy planning is still controversial [6,15,21,46,53]. Enucleation has been the conventional treatment of intraocular melanoma for at least 50

years but no follow-up study could show any clear survival benefit [4,10,37,55] even if enucleation was preceded by preoperative radiotherapy [11,16].

Radiotherapy with heavy ions, which is available only in very few medical centers, seems to be an effective treatment for uveal melanoma which combines high rates of local control, survival and eye retention with a substantial likelihood of long-term vision preservation [17,38,46].

However, application of radioactive plaques with isotopes such as cobalt-60 [54], iodine-125 or ruthenium-106 [39] is the primary treatment in most centers. In Germany, treatment with ruthenium-106 plaques has been favored in the recent years [20,33,40,43]. One radiotherapy session with radioactive applicators differs from the other and is never standar-

* Corresponding author.